Ultimate Bots robots at the Gemma Edge & Physical AI Build Day

AGI House · Friday, October 24, 2026 · Ultimate Bots table

The robots

Ultimate Bots brings four platforms:

- **Booster K1s** — the preferred robot and the most available one. 95 cm humanoid, Jetson inside, stereo camera, mic array, speaker, Python SDK, MuJoCo twin. The docs below are for the K1.
- **Unitree G1** — full-size humanoid.
- **Agibot X2** — full-size humanoid.
- **YAM Ultra arms** — a pair of manipulator arms.

The G1, X2 and the arms are on the floor with their own handlers — ask the handler for collaboration. Access to them is the handler's call; if you want one, you work as a team with the handler. All four exist in simulation, and the bigger humanoids and the arms are where some models fit best — so build and show it in sim, and the handler may put it on the real robot.

Robot access

Come to the Ultimate Bots table after the keynote with your team and your idea. How robot time is split depends on how many teams want a robot and for what; we will accommodate as many as we can and decide on the day. Either way: sim first; your code runs on your laptop or Jetson kit on the robot Wi-Fi and drives the robot through its SDK; Booster K1s can get dedicated access, other on-robot deployments are done with a handler.

What the K1 has

Size	95 cm, 19.5 kg
Joints	22 DoF — 6 per leg, 4 per arm, 2 head (yaw, pitch)
Compute	NVIDIA Jetson Orin NX 8 GB (117 TOPS), 512 GB NVMe
Vision	Wide-angle stereo camera in the head, 105° × 94° FOV
Audio	6-mic array in the head, speaker in the torso
Sensing	9-axis IMU, encoders and torque feedback on every joint
Battery	5 Ah, about 70 min walking; charges while

	running
Walking	Up to 2 m/s; capped at 0.5 m/s on the day
Network	Wi-Fi 6, gigabit Ethernet, Bluetooth 5.2
Software	Ubuntu 22.04, ROS 2 Humble, Booster SDK (Python, C++) over DDS

Gemma at the edge

Gemma 4 **E2B** and **E4B** take image and audio natively and do function calling — that is the whole robot brain. E2B in 4-bit should fit the K1's own Jetson (8 GB) — it was shown running fully local on an 8 GB Orin Nano with camera, mic and tool calls. E4B calls tools noticeably better; run it on a 16 GB Jetson or a laptop GPU on the robot Wi-Fi. Gemma answers in seconds, so let it decide at about 1 Hz and let the robot act — the K1 balances and walks on its own, you send velocity and gesture commands.

- Gemma 4 on Jetson — containers, GGUF and vLLM paths, tool-calling flags (NVIDIA Jetson AI Lab)
- Gemma 4 at the edge — NVIDIA's overview of the variants and what runs where
- Gemma 4 launch post — model cards, audio and vision encoders, function-calling format
- E2B on an 8 GB Orin Nano — Google's vision + voice + camera-control demo

The K1 SDK and the simulator

- Booster developer guide — start here
- BoosterOS Python SDK · quick start (pip install boosteros) · vision and speech — image, joints, IMU, mode, speech from Python
- Motion API — modes, walking, head, arm end-effectors, actions and sound · camera · audio
- booster_robotics_sdk — the C++ and Python bindings on GitHub
- K1 manual — modes · voice interaction
- Simulation: booster_assets (K1 URDF + MuJoCo XML) · booster_deploy (the robot's walking policy in MuJoCo, same command interface as hardware — guide) · booster_train (Isaac Lab, K1 motion tracking — not a one-day job)

The loop that works in a day: declare a handful of tools for Gemma (walk, turn, look, wave, say, stop) → Gemma returns a tool call → one SDK call each → a wrapper that clamps velocity and rate-limits. Build it against the MuJoCo K1, point it at the real one when you get the robot.

Ideas

- **Concierge** — "take me to the coffee": hear, plan, walk, look, wave on arrival.
- **Eyes** — scan the room, find the person in the red shirt, walk over, introduce itself.
- **Commentator** — watch the robot fight on the floor and call it live, in character.
- **Safety monitor** — camera and IMU in, narration out, hard stop when something looks wrong.
- **Character** — a personality, a voice, gestures, memory of who it met today.
- **Phone brain** — Gemma on a Pixel decides, the K1 acts.

- **Sim only** — Gemma composes a routine (walk, turn, wave, bow) in MuJoCo; the best ones run on the real K1 at demo time.

Before Friday

Download weights and containers, install the SDK, run the MuJoCo K1 once, bring an Ethernet dongle.